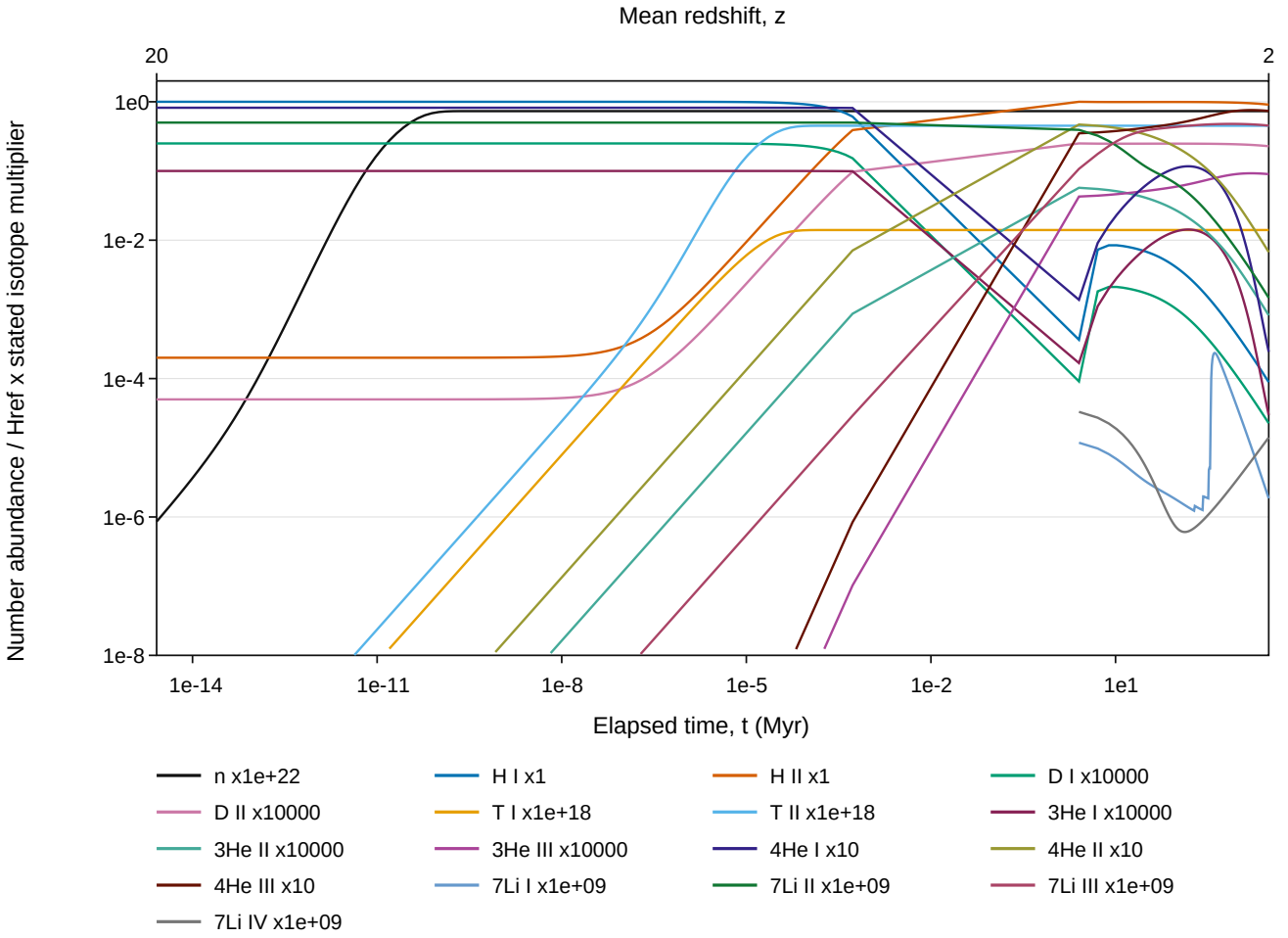
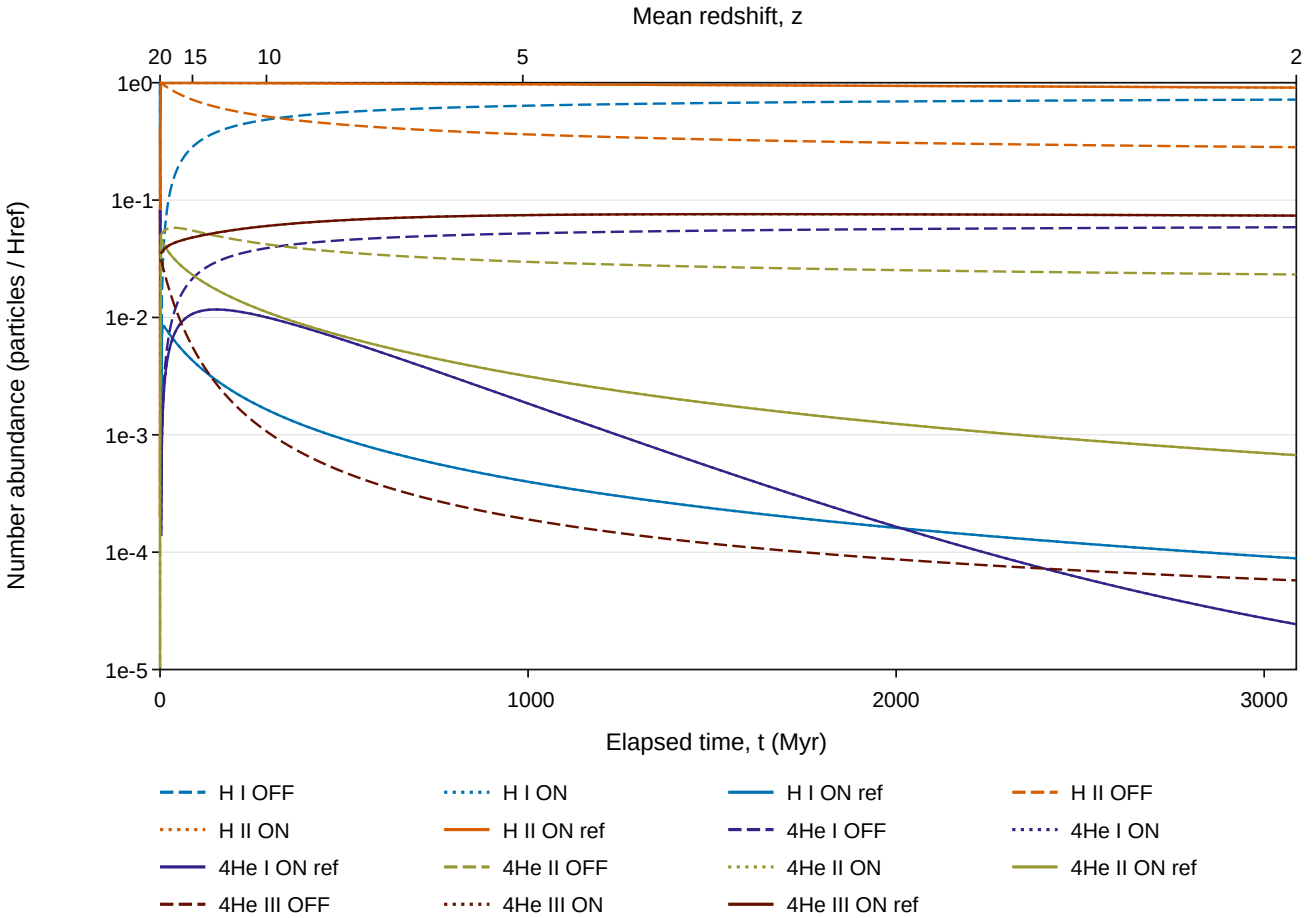
(a) All 17 states: ON_refined



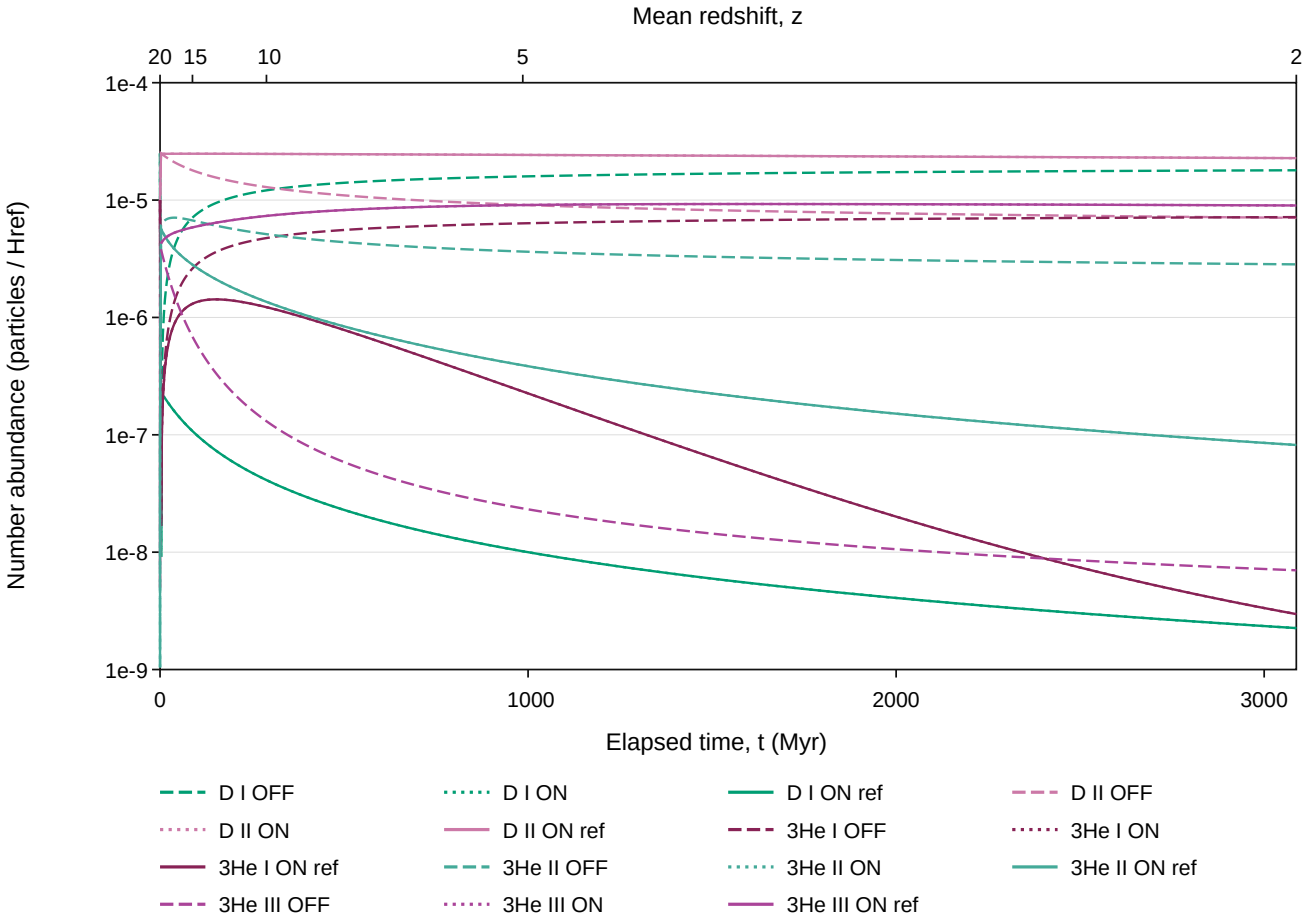
Fixed isotope multipliers appear in legend; y window = 1e-8 to 2.
Positive offscale counts by state are in FIGURE_METADATA.json.
t = 0 and nonpositive samples omitted; unscaled traces in grouped views.
Source: saved pilot17_populations.csv and three-case NPZ arrays.

(b) Abundance band: peak >= 1e-3



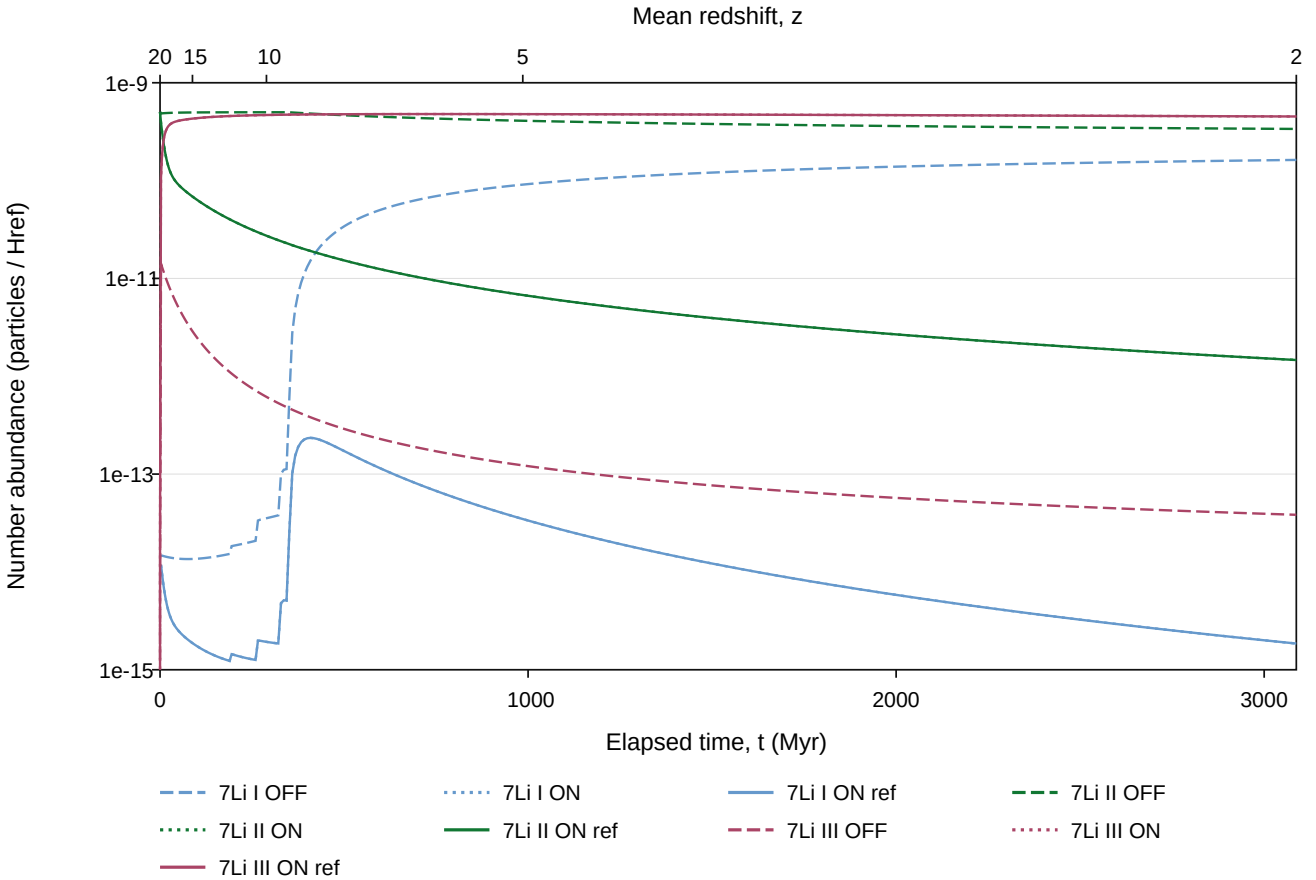
OFF dashed; ON_base dotted; ON_refined solid. State colors match overview.
Y limits: positive samples at t >= 1 Myr, rounded outward to decades.
Full time span retained; offscale counts per curve in FIGURE_METADATA.json.
Early source-onset evolution is resolved in figure (i).
Source: saved pilot17_populations.csv and three-case NPZ arrays.

(c) Abundance band: 1e-6 <= peak < 1e-3



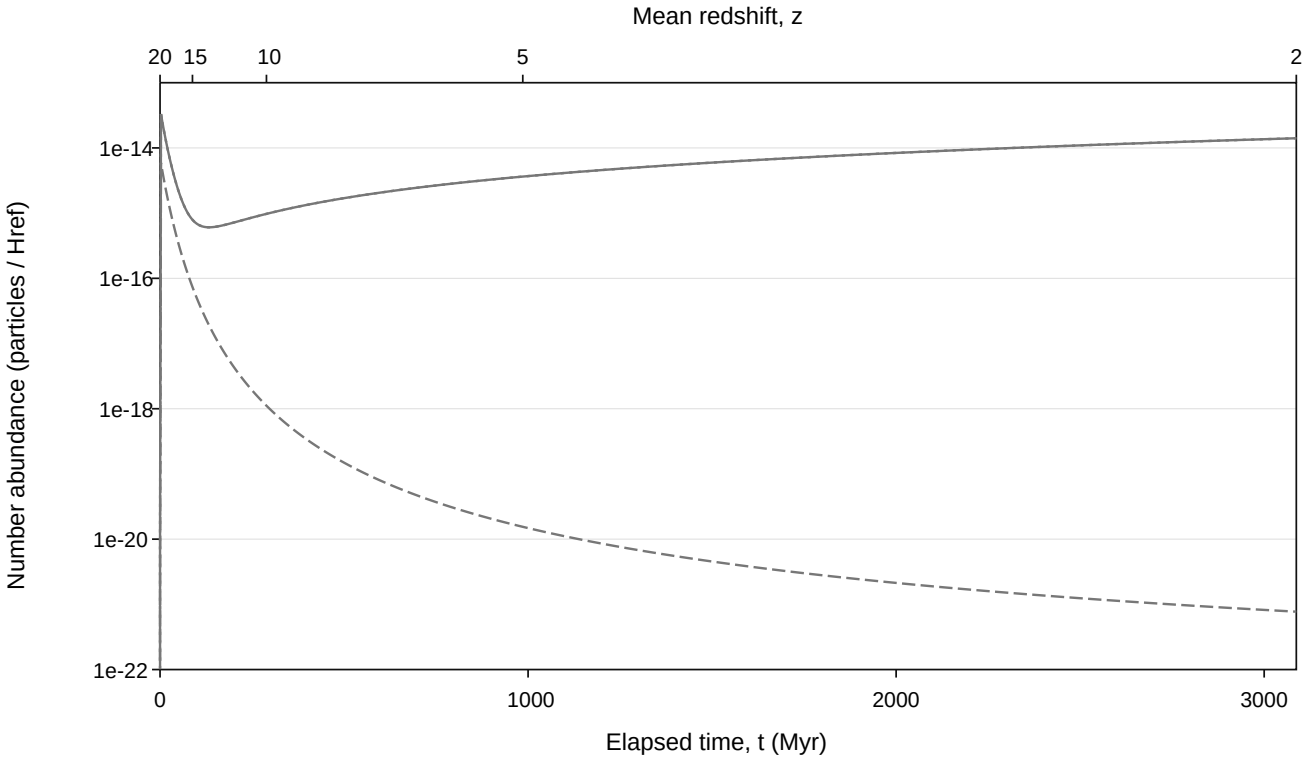
OFF dashed; ON_base dotted; ON_refined solid. State colors match overview.
Y limits: positive samples at t >= 1 Myr, rounded outward to decades.
Full time span retained; offscale counts per curve in FIGURE_METADATA.json.
Early source-onset evolution is resolved in figure (i).
Source: saved pilot17_populations.csv and three-case NPZ arrays.

(d) Abundance band: 1e-12 <= peak < 1e-6



OFF dashed; ON_base dotted; ON_refined solid. State colors match overview.
Y limits: positive samples at t >= 1 Myr, rounded outward to decades.
Full time span retained; offscale counts per curve in FIGURE_METADATA.json.
Early source-onset evolution is resolved in figure (i).
Source: saved pilot17_populations.csv and three-case NPZ arrays.

(e) Abundance band: 1e-16 <= peak < 1e-12



OFF dashed; ON_base dotted; ON_refined solid. State colors match overview.

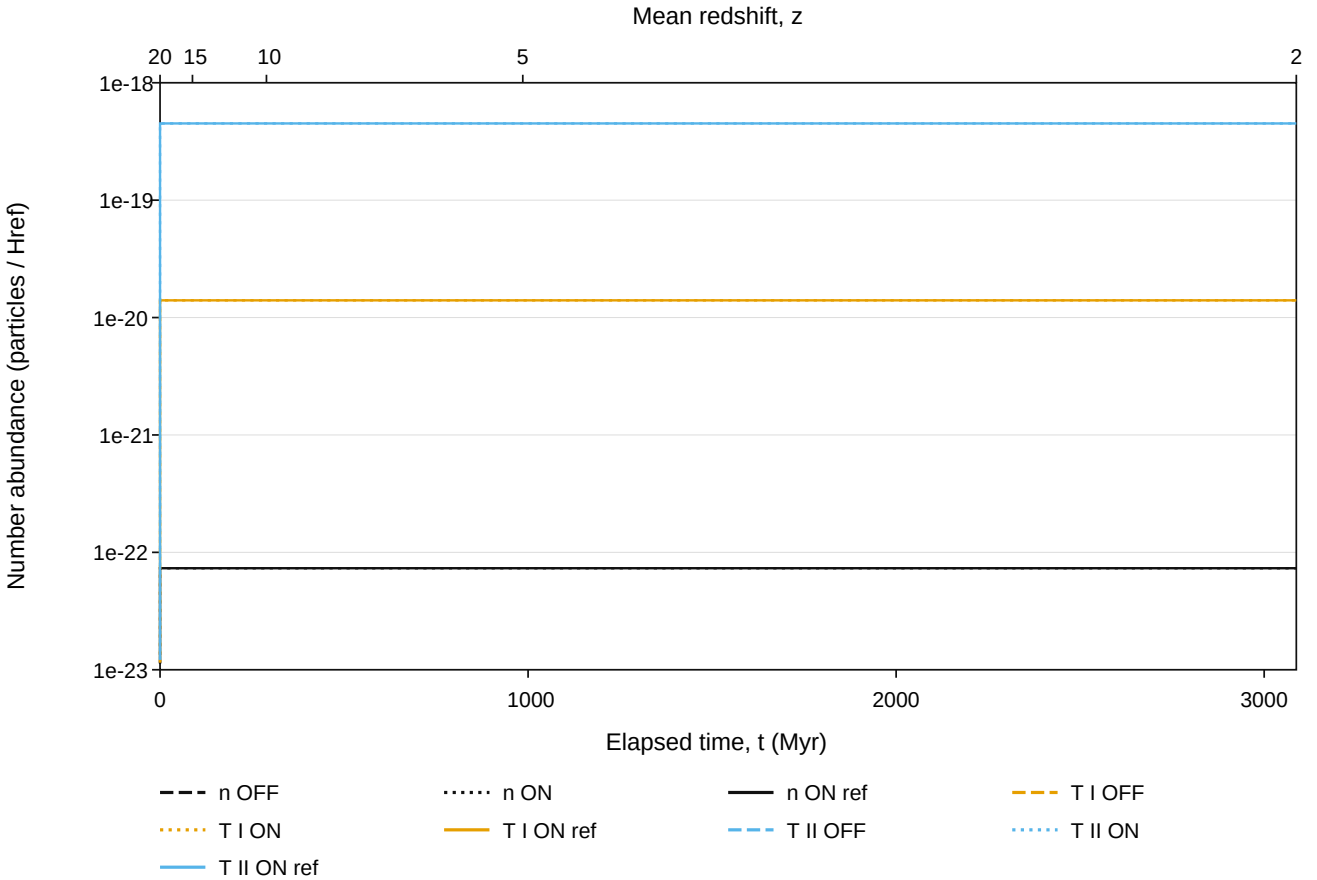
Y limits: positive samples at t >= 1 Myr, rounded outward to decades.

Full time span retained; offscale counts per curve in FIGURE_METADATA.json.

Early source-onset evolution is resolved in figure (i).

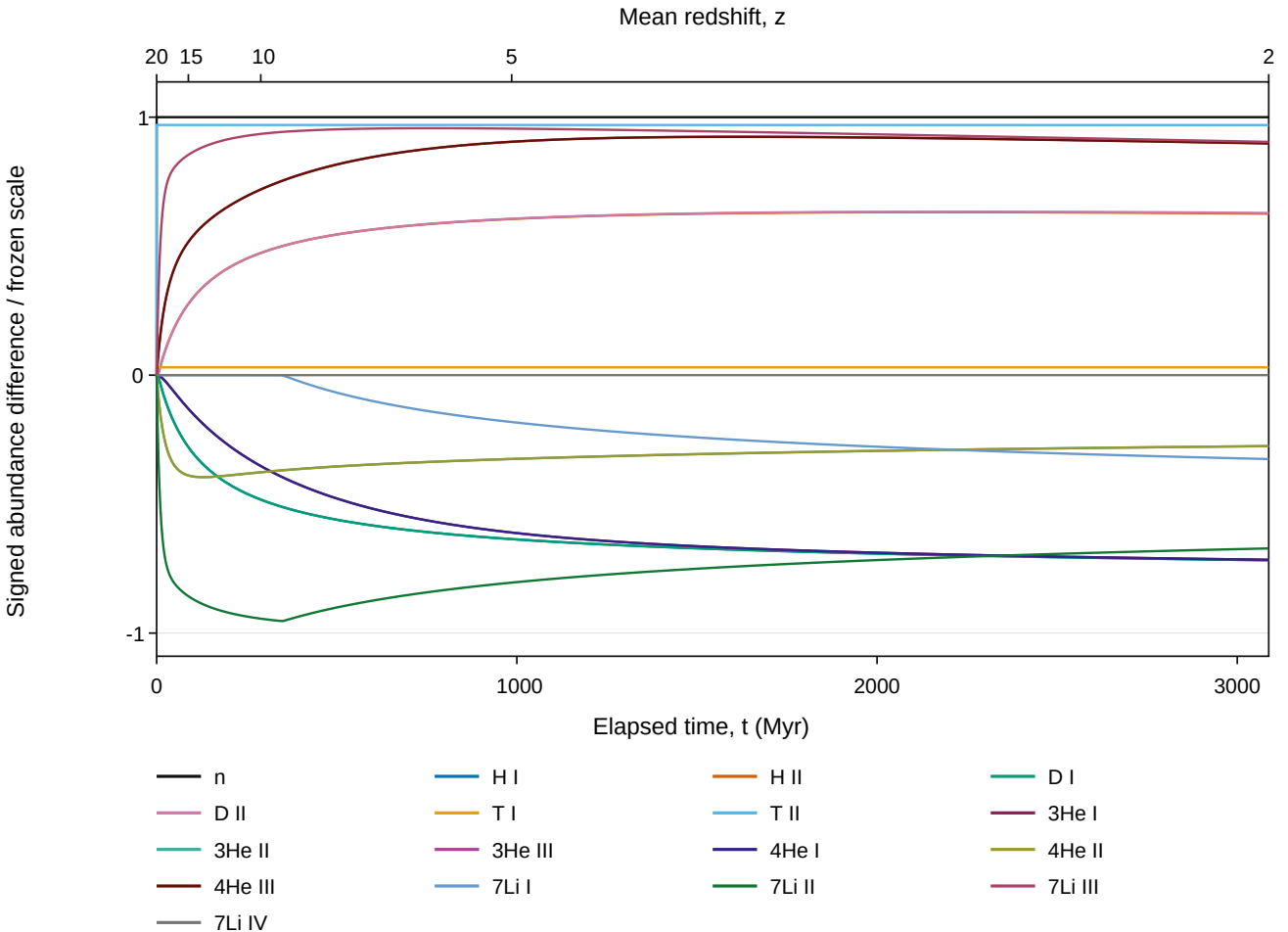
Source: saved pilot17_populations.csv and three-case NPZ arrays.

(f) Abundance band: peak < 1e-16



OFF dashed; ON_base dotted; ON_refined solid. State colors match overview.
Y limits: positive samples at t >= 1 Myr, rounded outward to decades.
Full time span retained; offscale counts per curve in FIGURE_METADATA.json.
Early source-onset evolution is resolved in figure (i).
Source: saved pilot17_populations.csv and three-case NPZ arrays.

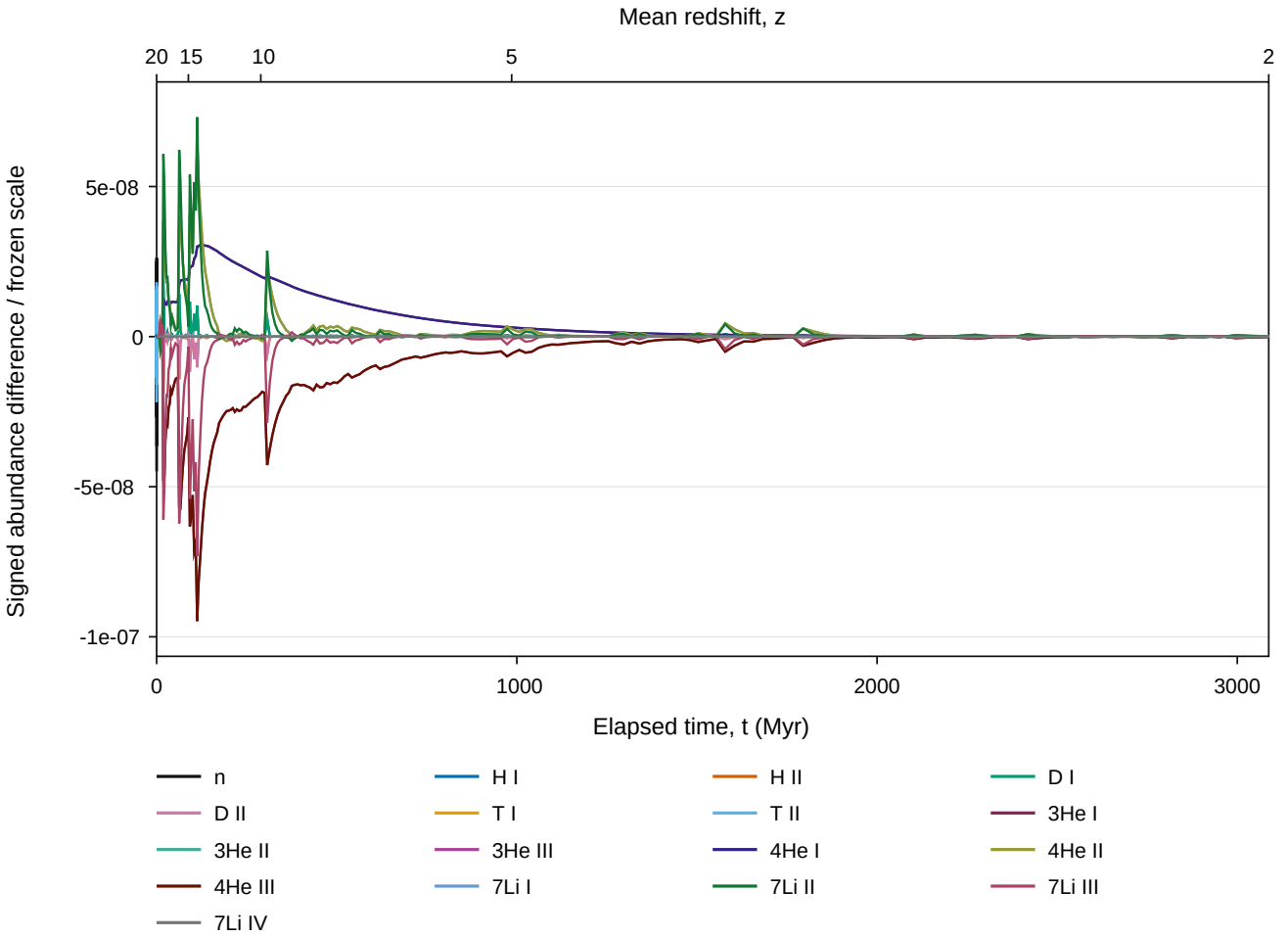
(g) Source response: ON_base - OFF_base



Full configured source switch, including external escape; zero retained.

Source: saved pilot17_populations.csv and three-case NPZ arrays.

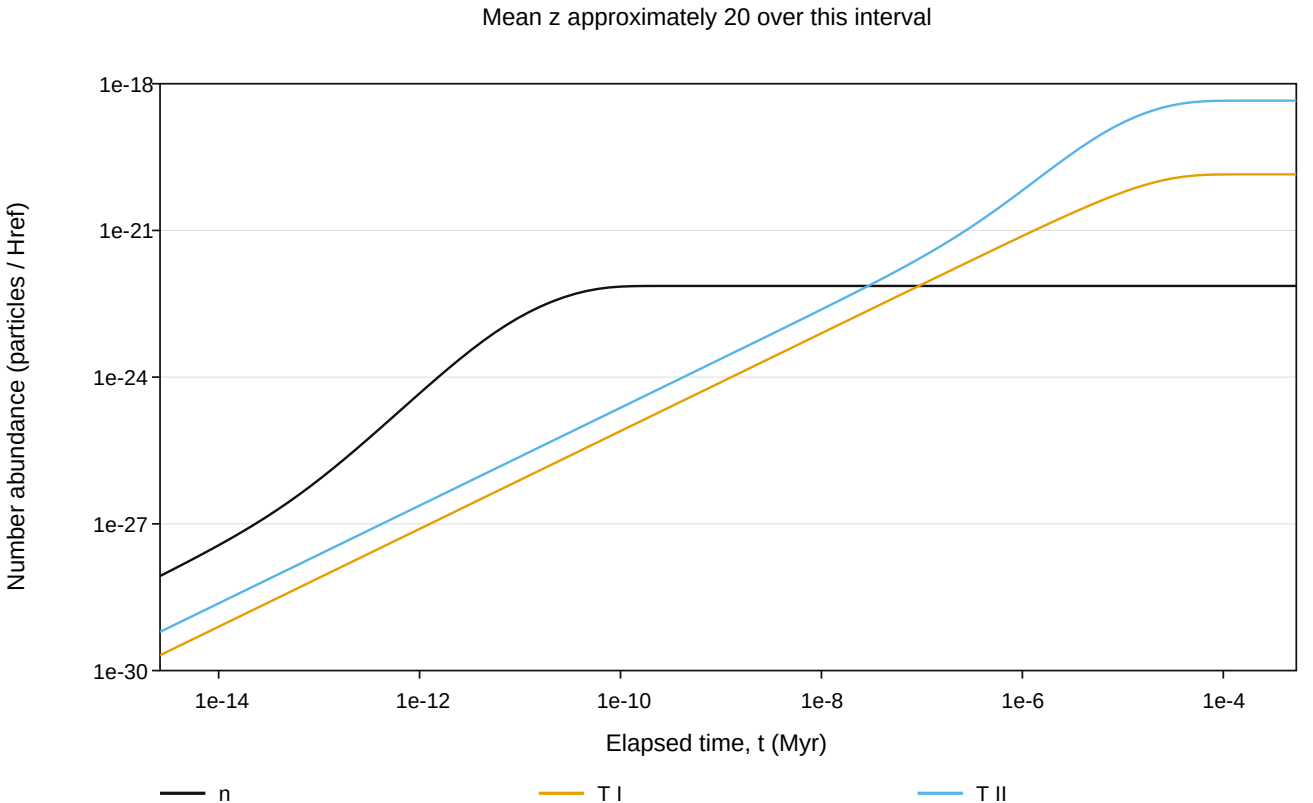
(h) Refinement: ON_base - ON_refined



Two-resolution agreement only; acceptance max |difference / scale| <= 1e-5.

Source: saved pilot17_populations.csv and three-case NPZ arrays.

(i) Initial transient: ON_refined

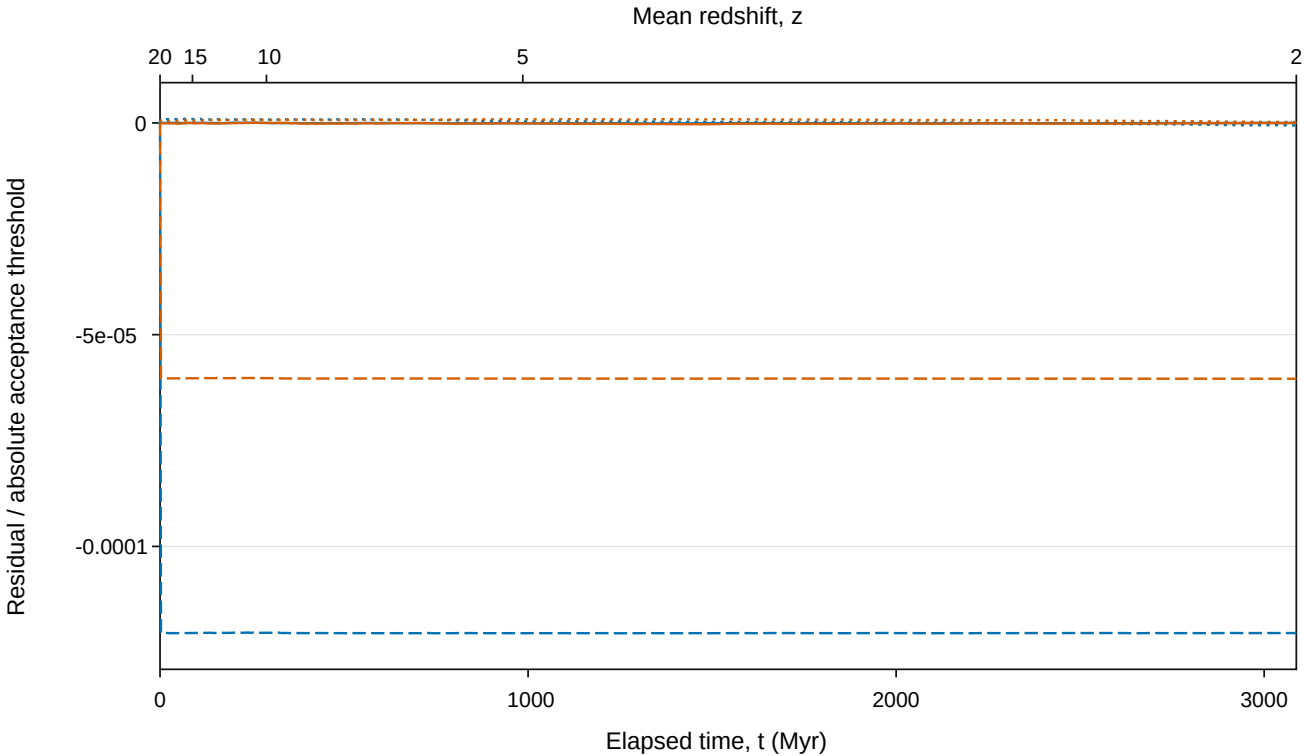


Source onset is t = 0; n, T I and T II begin at zero.

Positive saved samples only; no time relocation or abundance floor.

Source: saved pilot17_populations.csv and three-case NPZ arrays.

(j) Signed open-system conservation residuals



Baryon + external ledger; heavy charge - electron ledger; initial values removed.

Original absolute threshold = 1.0820350005000101e-8 particles / Href.

Source: saved pilot17_populations.csv and three-case NPZ arrays.